

# Christian Gonzalez-Ruiz

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## EDUCATION

**University of California, Santa Barbara**

Jun 2025

*B.S. Mechanical Engineering*

## SKILLS

CAD/PLM: SolidWorks, NX | GD&T, drawings | CAE: ANSYS (static, modal, vibration), COMSOL (CFD/FSI)

Manufacturing: CNC machining, weldments, additive manufacturing (SLA/FDM/PBF) | Programming: Python, MATLAB

## EXPERIENCE

**Hyllion, Inc. | Milford, OH**

Aug 4th, 2025 – Present

*Mechanical Engineer*

- Led concept-to-review structural development for containerized energy generator systems (2 Megawatt concept design review; 800 kilowatt detailed design review), supporting first prototype build targeted for Q2 & Q3 2026.
- Performed frame architecture weld sizing and ANSYS FEA (static, modal, buckling) for transport and operational load cases (5 g transport; designed to 3 g operational acceleration), eliminating ~40 parts (~7% of ~600), reducing primary C-channel thickness from 3/8 in to 1/8 in where margins allowed, and increasing minimum natural frequency from 8 Hz to 31 Hz through X-bracing and gusseting to meet frequency-avoidance requirements.
- Presented analysis results and trade studies in concept/design reviews; documented assumptions, margins, and requirements compliance for cross-functional alignment. Partnered with suppliers on lead times and released drawings, cut lists, and material specs.
- Built Python automation for engineering calculations and NX geometry slicing for PBF, cutting build-prep time from ~1 hour to ~5 minutes.
- Currently designing LV cabinet thermal management to meet max component temperature limits; selecting fans/filters within a constrained envelope, and will validate airflow via CFD. The goal is to reduce the average internal temperature from ~200 C to ~55 C and verify structural integrity.
- Owning system-level subsystem re-layout, redesigning component placement and orientation (pumps, inverters, coolant tubing, and wiring) to eliminate air entrapment in coolant paths, while meeting envelope, serviceability, and manufacturability constraints; followed by structural FEA validation.

**Capella Partners | Santa Barbara, CA**

Aug 2024 – Jun 2025

*Mechanical Engineer Intern*

- Designed and updated 300+ part CAD assemblies for a hydrogen generation system; created engineering drawings for manufacturers.
- Fabricated and tested parts in Al 6061-T6, PEEK, and stainless steels using CNC machining and 3D printing (SLA/FDM); supported lab experiments and rapid iteration.
- Built fluid-path demonstration prototypes used for external presentations to stakeholders/investors.

**Vascular BioSciences | Santa Barbara, CA**

Jan 2024 – Jun 2025

*Mechanical / Manufacturing Engineer Intern*

- Prototyped medical devices using CAD and 3D printing; supported DFM improvements and precision measurement workflows.
- Troubleshoot rotary evaporators and vacuum systems, cutting drug development cycle time from 48 hours to 4 hours (83%) using fluid dynamics principles and root-cause analysis (5 Whys).
- Performed cleanroom testing and documentation of catheter fabrication to verify quality and process consistency.

## PROJECTS

**Cadense, Inc. – Hypershoe Optimization (Capstone, Most Technical Project Award)**

Sep 2024 – Jun 2025

- Ran solid mechanics, CFD, and fluid-structure interaction simulations in COMSOL to evaluate stress, temperature distribution, and flow velocities for energy absorption optimization.
- Designed and built an impact testing machine (machined/welded) for prototype benchmarking; produced GD&T drawings for fabrication.